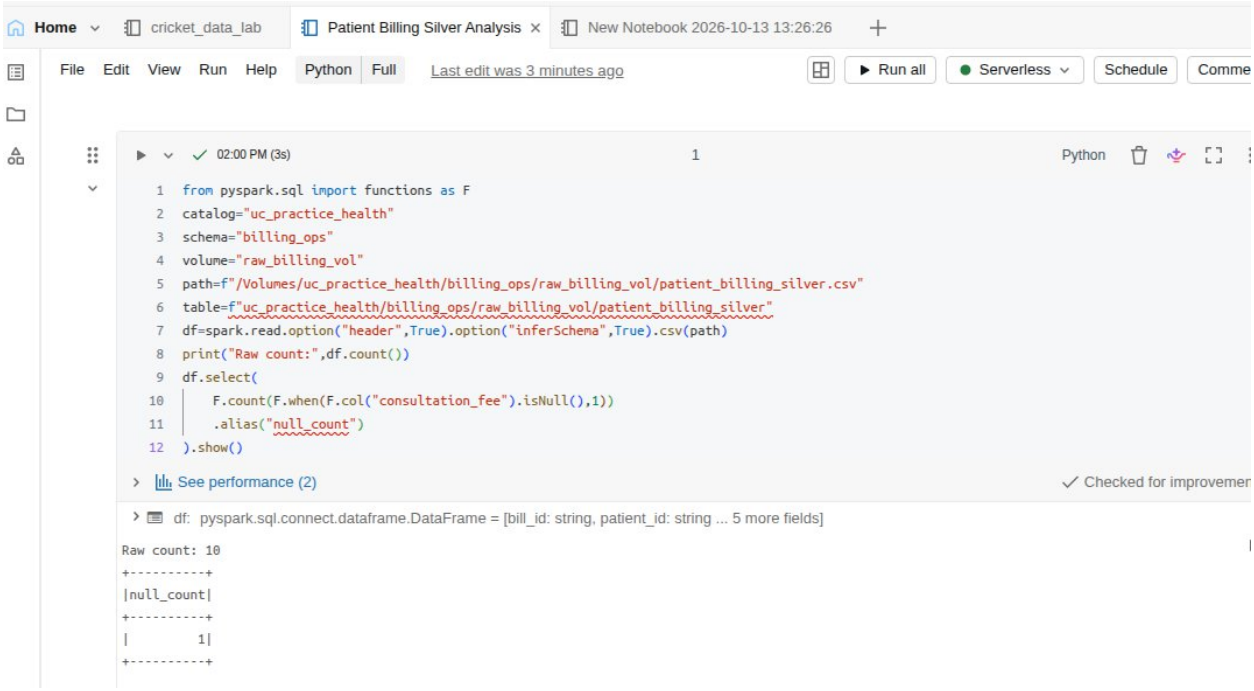
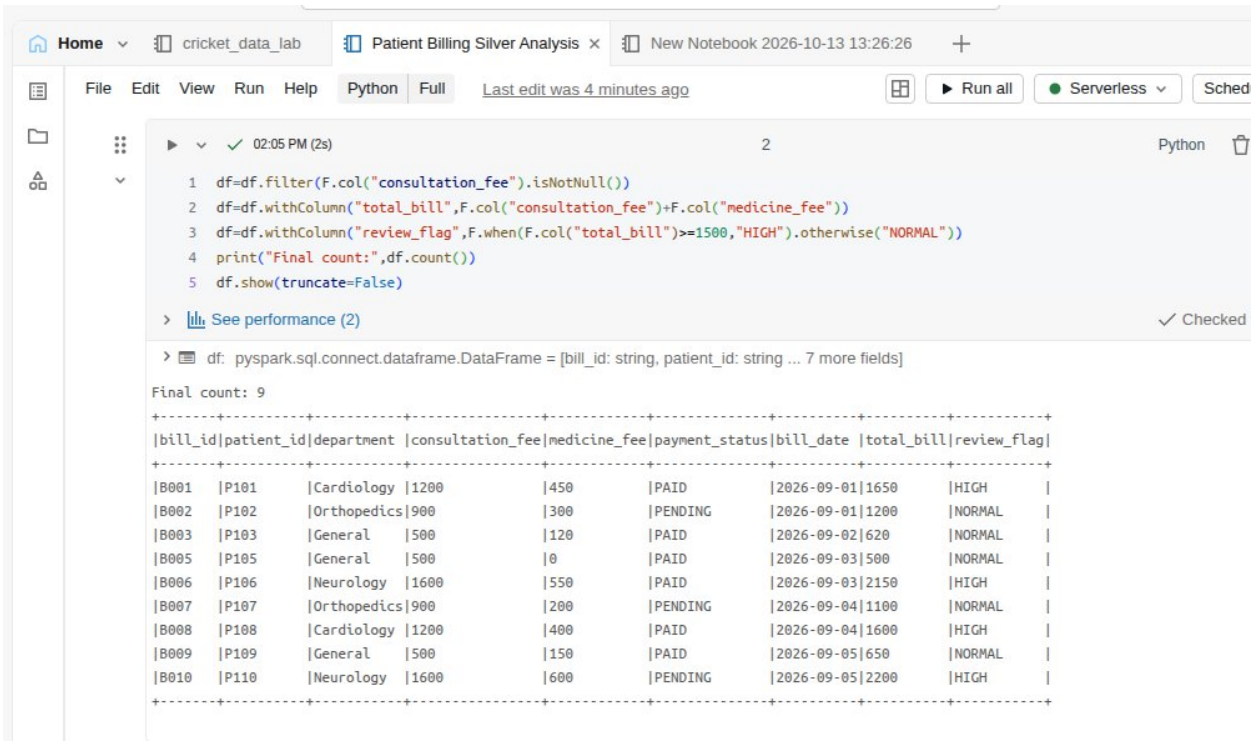
WEEK 9 DAY 2 COD ASSIGNMENT TASK OUTPUTS :

QUESTION 1 : HEALTHCARE PATIENT BILLING



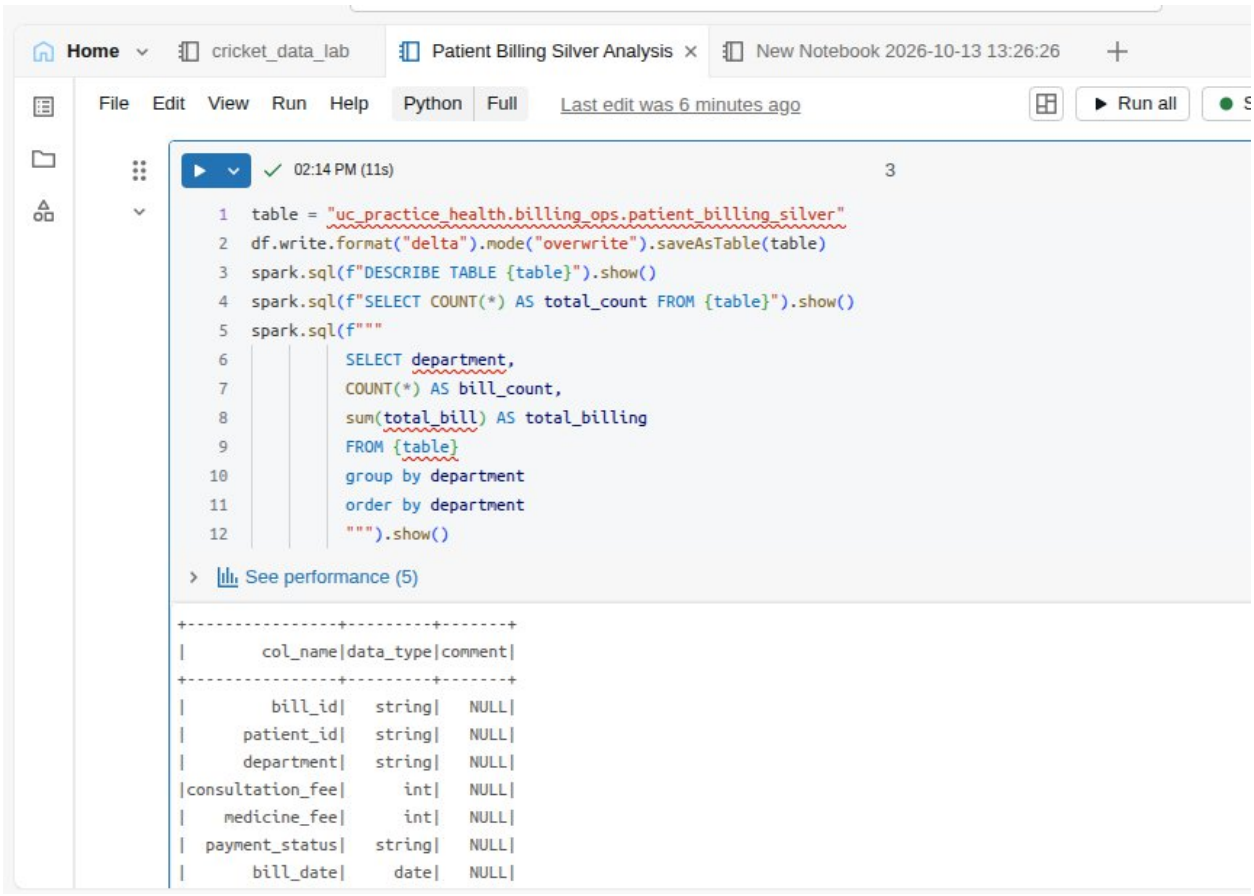
```
from pyspark.sql import functions as F
catalog="uc_practice_health"
schema="billing_ops"
volume="raw_billing_vol"
path=f"/Volumes/uc_practice_health/billing_ops/raw_billing_vol/patient_billing_silver.csv"
table=f"uc_practice_health/billing_ops/raw_billing_vol/patient_billing_silver"
df=spark.read.option("header",True).option("inferSchema",True).csv(path)
print("Raw count:",df.count())
df.select(
    F.count(F.when(F.col("consultation_fee").isNull(),1))
    .alias("null_count")
).show()
```

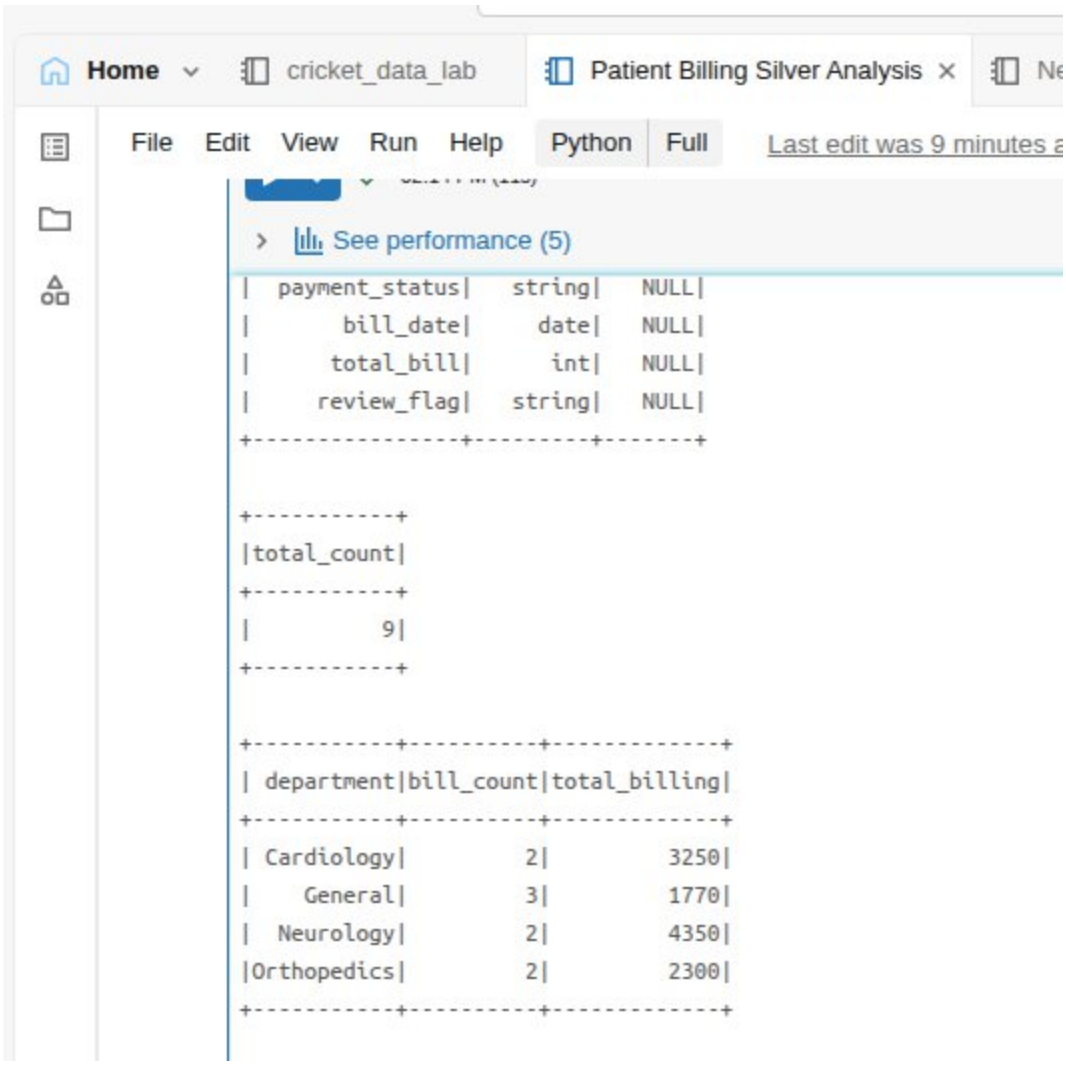
```
Raw count: 10
+----------+
|null_count|
+----------+
|         1|
+----------+
```

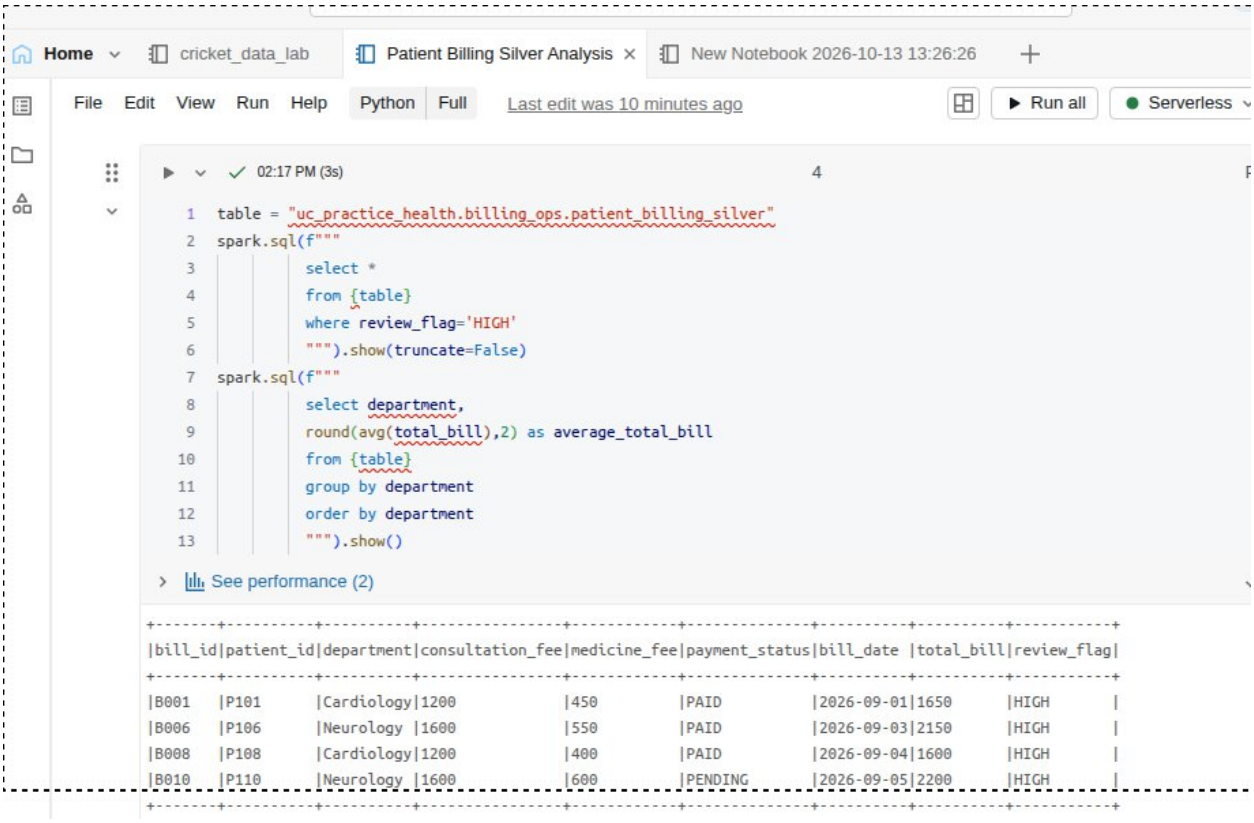


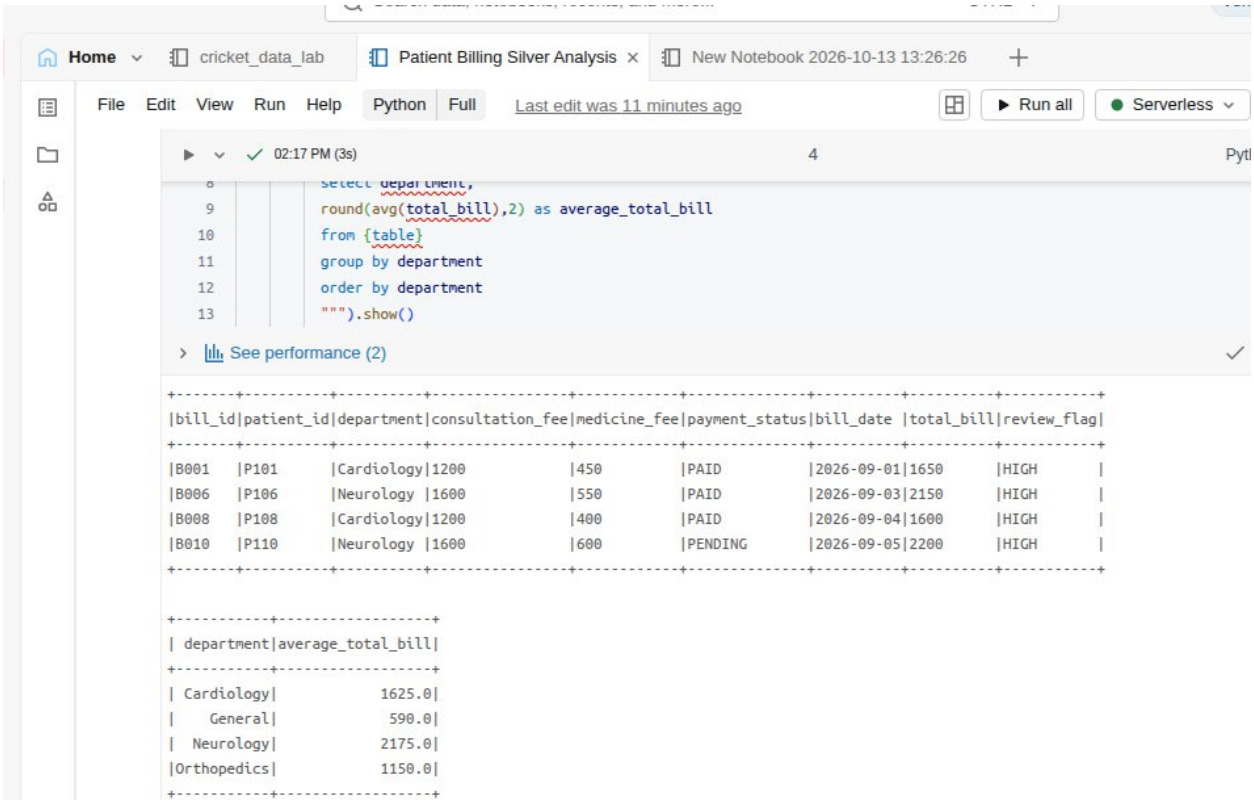
```
df=df.filter(F.col("consultation_fee").isNotNull())
df=df.withColumn("total_bill",F.col("consultation_fee")+F.col("medicine_fee"))
df=df.withColumn("review_flag",F.when(F.col("total_bill")>=1500,"HIGH").otherwise("NORMAL"))
print("Final count:",df.count())
df.show(truncate=False)
```

```
Final count: 9
+-------+----------+-----------+----------------+------------+--------------+----------+----------+-----------+
|bill_id|patient_id|department |consultation_fee|medicine_fee|payment_status|bill_date |total_bill|review_flag|
+-------+----------+-----------+----------------+------------+--------------+----------+----------+-----------+
|B001   |P101      |Cardiology |1200            |450         |PAID          |2026-09-01|1650      |HIGH       |
|B002   |P102      |Orthopedics|900             |300         |PENDING       |2026-09-01|1200      |NORMAL     |
|B003   |P103      |General    |500             |120         |PAID          |2026-09-02|620       |NORMAL     |
|B005   |P105      |General    |500             |0           |PAID          |2026-09-03|500       |NORMAL     |
|B006   |P106      |Neurology  |1600            |550         |PAID          |2026-09-03|2150      |HIGH       |
|B007   |P107      |Orthopedics|900             |200         |PENDING       |2026-09-04|1100      |NORMAL     |
|B008   |P108      |Cardiology |1200            |400         |PAID          |2026-09-04|1600      |HIGH       |
|B009   |P109      |General    |500             |150         |PAID          |2026-09-05|650       |NORMAL     |
|B010   |P110      |Neurology  |1600            |600         |PENDING       |2026-09-05|2200      |HIGH       |
+-------+----------+-----------+----------------+------------+--------------+----------+----------+-----------+
```


Home
cricket_data_lab
Patient Billing Silver Analysis
New Notebook 2026-10-13 13:26:26
File Edit View Run Help
Python Full
Last edit was 6 minutes ago
Run all
02:14 PM (11s)
3
1 table = "uc_practice_health.billing_ops.patient_billing_silver"
2 df.write.format("delta").mode("overwrite").saveAsTable(table)
3 spark.sql(f"DESCRIBE TABLE {table}").show()
4 spark.sql(f"SELECT COUNT(*) AS total_count FROM {table}").show()
5 spark.sql(f"""
6 SELECT department,
7 COUNT(*) AS bill_count,
8 sum(total_bill) AS total_billing
9 FROM {table}
10 group by department
11 order by department
12 """).show()
See performance (5)
+----------+---------+-------+
| col_name|data_type|comment|
+----------+---------+-------+
bill_id	string	NULL
patient_id	string	NULL
department	string	NULL
consultation_fee	int	NULL
medicine_fee	int	NULL
payment_status	string	NULL
bill_date	date	NULL


Home
cricket_data_lab
Patient Billing Silver Analysis
File Edit View Run Help
Python
Full
Last edit was 9 minutes a
See performance (5)
payment_status	string	NULL
bill_date	date	NULL
total_bill	int	NULL
review_flag	string	NULL
+----------+---------+-------+
+-----------+
|total_count|
+-----------+
| 9|
+-----------+
+----------+----------+-------------+
| department|bill_count|total_billing|
+----------+----------+-------------+
Cardiology	2	3250
General	3	1770
Neurology	2	4350
Orthopedics	2	2300
+----------+----------+-------------+


Home
cricket_data_lab
Patient Billing Silver Analysis
New Notebook 2026-10-13 13:26:26
File Edit View Run Help Python Full
Last edit was 10 minutes ago
Run all
Serverless
02:17 PM (3s)
4
table = "uc_practice_health.billing_ops.patient_billing_silver"
spark.sql(f"""
select *
from {table}
where review_flag='HIGH'
""").show(truncate=False)
spark.sql(f"""
select department,
round(avg(total_bill),2) as average_total_bill
from {table}
group by department
order by department
""").show()
See performance (2)
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+
|bill_id|patient_id|department|consultation_fee|medicine_fee|payment_status|bill_date |total_bill|review_flag|
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+
B001	P101	Cardiology	1200	450	PAID	2026-09-01	1650	HIGH
B006	P106	Neurology	1600	550	PAID	2026-09-03	2150	HIGH
B008	P108	Cardiology	1200	400	PAID	2026-09-04	1600	HIGH
B010	P110	Neurology	1600	600	PENDING	2026-09-05	2200	HIGH
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+


Home
cricket_data_lab
Patient Billing Silver Analysis
New Notebook 2026-10-13 13:26:26
File Edit View Run Help Python Full
Last edit was 11 minutes ago
Run all
Serverless
02:17 PM (3s)
4
round(avg(total_bill),2) as average_total_bill
from {table}
group by department
order by department
""").show()
See performance (2)
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+
|bill_id|patient_id|department|consultation_fee|medicine_fee|payment_status|bill_date |total_bill|review_flag|
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+
B001	P101	Cardiology	1200	450	PAID	2026-09-01	1650	HIGH
B006	P106	Neurology	1600	550	PAID	2026-09-03	2150	HIGH
B008	P108	Cardiology	1200	400	PAID	2026-09-04	1600	HIGH
B010	P110	Neurology	1600	600	PENDING	2026-09-05	2200	HIGH
+-------+----------+----------+----------------+------------+--------------+----------+----------+-----------+
+-----------+------------------+
| department|average_total_bill|
+-----------+------------------+
Cardiology	1625.0
General	590.0
Neurology	2175.0
Orthopedics	1150.0
+-----------+------------------+